1. **Find the IP address of your computer or mobile device and write down whether it is IPv4 or IPv6. Also mention if it is a public or private IP.**

```
Wireless LAN adapter WiFi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2001:db8:1::100
   IPv6 Address. . . . . . . . . . . : 2402:a00:408:8f67:7ec3:d8aa:6308:d824
   IPv6 Address. . . . . . . . . . . : fd00:1:1::100
   Temporary IPv6 Address. . . . . . : 2402:a00:408:8f67:85a2:2381:cdf2:43a0
   Link-local IPv6 Address . . . . . : fe80::bcff:c2db:3220:39b7%6
   IPv4 Address. . . . . . . . . . . : 192.168.1.72
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1%6
                                       192.168.1.254
```

Here, my laptop's IP address is 192.168.1.72 which is IPv4 private IP address.

2. **Given the following list of IP addresses, identify which ones are valid private IPs and which are public: 192.168.1.10, 10.0.0.5, 172.16.5.20, 8.8.8.8, 123.45.67.89.**

- The standard private IPv4 address ranges are:
- Class-A 10.0.0.0 – 10.255.255.255
- Class-B 172.16.0.0 – 172.31.255.255
- Class-C 192.168.0.0 – 192.168.255.255

IP Address	**Type**	**Reason**
192.168.1.10	Private	Falls within 192.168.0.0/16
10.0.0.5	Private	Falls within 10.0.0.0/8
172.16.5.20	Private	Falls within 172.16.0.0/12
8.8.8.8	Public	Not in any private range
123.45.67.89	Public	Not in any private range

Here, Private IPs is 192.168.1.10, 10.0.0.5 and 172.16.5.20

And Public IPs is 8.8.8.8 and 123.45.67.89

3. **Write a short script in Python or JavaScript that takes an IP address as input and tells the user whether it is a valid IPv4 address or not.**

```
Ip = input(“Enter an IP address: “)

Parts = ip.split(“.”)

If len(parts) == 4:

    Valid = True

    For part in parts:

        If not part.isdigit() or not 0 <= int(part) <=255:

            Valid = False

            Break

    If valid:

        Print(“Valid IPv4 address”)

    Else:

        Print(“Invalid IPv4 address”)

Else:

    Print(“Invalid IPv4 address”)
```

Example

- When I enter 192.168.1.10 it will give me a result like, “Valid IPv4 address”.
- Whereas when I enter 192.168.1.300 it will give me a result like “Invalid IPv4 address”.

4. **Explain with an example how Instagram or WhatsApp might use IP addresses to detect suspicious login activity from different locations.**

- Instagram or WhatsApp can use the IP address associated with a login to estimate where the login is coming from. They can compare it with user's previous login patterns to identify unusual activity.

Example

- I normally use WhatsApp and Instagram from Ahmedabad, India.

Login	**Approx location**	**Result**
Monday, 9:00AM	Ahmedabad, India	Normal
Tuesday, 8:30PM	Ahmedabad, India	Normal
Wednesday, 10:10AM	London, UK	Suspicious

- If the service sees a login from an IP address that appears to be in London shortly after normal Ahmedabad login. It may consider this unusual.
- The system might then, check the IP address and approximate location.
- Compare it with previous login locations and devices.
- Check other signals such as whether the device is new or whether many login attempts occurred.
- If the activity looks suspicious, it could trigger additional verification such as an OTP or security notification.
- You might receive a message such as "Was this you?" or be asked to confirm the login.